

SATELLITES Deciphered



Mahesh K.C.
is analogue
applications
engineer
at Texas
Instruments



Srijan Ashok
is technical
sales engineer
at Texas
Instruments

We are nothing but a speck on the canvas of the universe; but this has not set a limit to our ambitions of exploring it. Space has always managed to fuel our interest from time immemorial and instigated us to continuously pursue it.

In this article we intend to introduce the basics of a satellite with respect to the electronics used in internal systems that is necessary to meet the purpose of a satellite (or payload). Before we actually move on to introducing satellites, let us first understand the challenges that Space has to offer.

Space is generally considered devoid of any medium. Here, the only means of energy transfer is through radiation. Conventional heat dissipation through convection and conduction, which characterises junction thermal resistance on Earth, is not applicable. New thermal models have to be created to analyse thermal performance.

A satellite orbiting in Space is showered with radiations from many sources, mainly the Sun. The Sun emits charged particles like alpha and beta that strike the satellite, causing transient effects in semiconductors. Uncharged gamma and ultraviolet rays hit the lattice of the semiconductor and disrupt its structure.

In spite of all these, the satellite has to still survive and operate. In order to describe this harsh environment and esti-

mate the lifetime of satellites, we quantify these effects mainly into total ionisation dose (TID) and single-event effects (SEEs).

TID is defined as the cumulative amount of radiation energy that the semiconductor can handle before it fails. It is measured in Rad (unit of radiation).

SEEs like latchup, upset, bit flip, gate rupture and transients are measured in mega electron volts (MeV), which signifies the radiation energy required to cause SEE.

Semiconductors for Space undergo a special process called radiation hardening in order to attain a higher immunity to radiation effects. The extent of hardening depends on the orbit and criticality of the mission. A common number, referred to as a rule of thumb, for TID is 100kRad and 100MeV for SEE for a deep-Space mission.

System designs in the satellite can be classified into mission-critical and non-mission-critical. Mission-critical applications include those designs whose malfunction can cause an irreversible damage to the satellite and may also result in mission failure. These include sub-systems like attitude determination and control system (ADCS), onboard control and data handling (OBC&DH), electronic power system (EPS) and actuator electronics. These are explained in detail in subsequent sections.

Non-mission-critical designs typically include systems that are not continuously used throughout the life span of the satellite. These include payload sub-systems like cameras and imaging devices. The launch vehicle can also be put under this category as its utility ceases after it has put the satellite into orbit and is no longer needed for satellite operation. Damage to the satellite from these non-mission-critical designs is minimal.

With this basic understanding of Space and systems, we will study more about satellites. A satellite is defined as anything that orbits a planet. Satellites can be mainly categorised as:

Low-Earth orbit. These fly at low alti-

Fig. 1: Satellite sub-systems

